

POWERFACTORY

PowerFactory 2024

Technical Reference

DIgSILENT Grid-forming Converter Templates

Droop Controlled Converter,
Synchronverter,
Virtual Synchronous Machine

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 Introduction

Grid-forming controllers have been introduced to enhance the stability of power systems with a large share of generators that are interfaced via power electronic converters. Usually voltage-sourced converters (VSC) are used as grid interface of those generating units. VSC have traditionally been equipped with fast current controllers in the inner control loop, which make the converters behave similar to a controlled current source. This current source behaviour can cause stability issues in weak grids or grids with a high penetration of power electronic interfaced generators. Grid-forming controllers enable voltage source behaviour with inertia in the voltage angle to overcome the issues.

This document describes three common controller types, which are implemented as DSL models and provided as templates in the *DIgSILENT* library:

- the Virtual Synchronous Machine (VSM) (see Section 2.1),
- the Synchronverter (see Section 2.2) and
- droop control (see Section 2.3).

Grid-forming controllers can be used in combination with a virtual impedance which improves the damping and allows for current limiting in the event of short-circuits [2] [4]. The algebraic type virtual impedance is outlined in Section 2.4. In addition, the block for output voltage calculation includes further functionality for current limitation (see Section 2.5). The grid-forming converter controls introduced in the described model templates are designed to not contain further inner loop controls. Therefore any further possible inner loop controls, which are under investigation by academia, and which can be used for further current limitation, are not part of the described models. Note that there is currently no means of current limitation in lower-level controls (such as current clipping) implemented in the models.

The document is structured as follows: In Sec. 2 the theory behind the grid-forming controller types is elaborated and the structure of the DSL models is depicted. Sec. 3 and 4 discuss particularities of RMS- and EMT-Simulations, respectively.

2 General Description

2.1 Virtual Synchronous Machine

The composite model of the Virtual Synchronous Machine (VSM) is shown in Fig. 2.1. It comprises the current and voltage measurement, the power calculation, the grid-forming controller (with the actual VSM), the virtual impedance, the virtual impedance control, the voltage controller, the output voltage calculation and a slot for a static generator, which is the controlled network element. The virtual impedance is covered in Sec. 2.4.
The active and reactive power are calculated as follows:

$$p_{mea} = u_r i_r + u_i i_i, \tag{1}$$

$$q_{mea} = u_i i_r - u_r i_i. \tag{2}$$

Note that only the positive sequence is considered in RMS-simulations.
The VSM is a type of grid-forming control that emulates the mechanical part of a synchronous machine, which is also referred to as the swing equation. A VSM can be represented by the approximated synchronous machine per unit balance in the Laplace domain [1]:

$$T_a s\omega = p_{set} - p_{mea} - D_p(\omega_r - \omega_{set}), \tag{3}$$

where T_a is the mechanical time constant, p_{set} is the active power set point and p_{mea} is the measured active power output. The rotating speed of the VSM is given by ω_r, ω_{set} is the frequency set point and D_p is the damping coefficient.
The VSM controller is illustrated in Fig 2.2. The switches are in position 1. The lower part of the Fig. 2.2 has no influence on the VSM and is only relevant for the Synchronverter. In addition to the swing equation, a low-pass filter ($\frac{\omega_c}{\omega_c+s}$) is applied in the damping term. This allows for inertia provision during transients, while eliminating the impact of the damping term at steady-state, i.e. eliminating the droop. If this steady-state impact is wanted, ω_c must be set to zero. The mechanical time constant T_a of the VSM can also be controlled by an input signal (not shown in Fig. 2.2).
Variable f_{nom} is the nominal frequency and f_{ref} is the reference frequency in RMS-simulations. Variable θ_v is the output voltage angle.
The VSM concept entails no further specification for the voltage magnitude control. The voltage control is implemented as a simple first order lag. More sophisticated voltage controllers can be added to the voltage control slot in Fig. 2.1.

2.2 Synchronverter

The Synchronverter was introduced in [5]. Its composite model is illustrated in Fig. 2.3. The measurement slots, the virtual impedance with its controller, the output voltage calculation and slot for the static generator are similar to the VSM. In addition, the current conversion slot converts the measured current into the phase currents (for EMT) or magnitude and angle (for RMS). There is no voltage control slot, as voltage control is integrated in the Synchronverter. The virtual impedance is elaborated in Sec. 2.4.
The Synchronverter control is shown in Fig. 2.2, the switches are in position 2. Similar to the VSM, it is also based on the swing equation. Hence, (3) also holds, except that the torques t_{set} and t_{calc} are used instead of p_{set} and p_{mea}. The main differences to the VSM are the additional reactive power control and the controller output voltage being used for the power/torque calculation instead of the measured voltage.
Eq. (7), (8) and (9) from [5] are at the core of the Synchronverter concept and calculate the

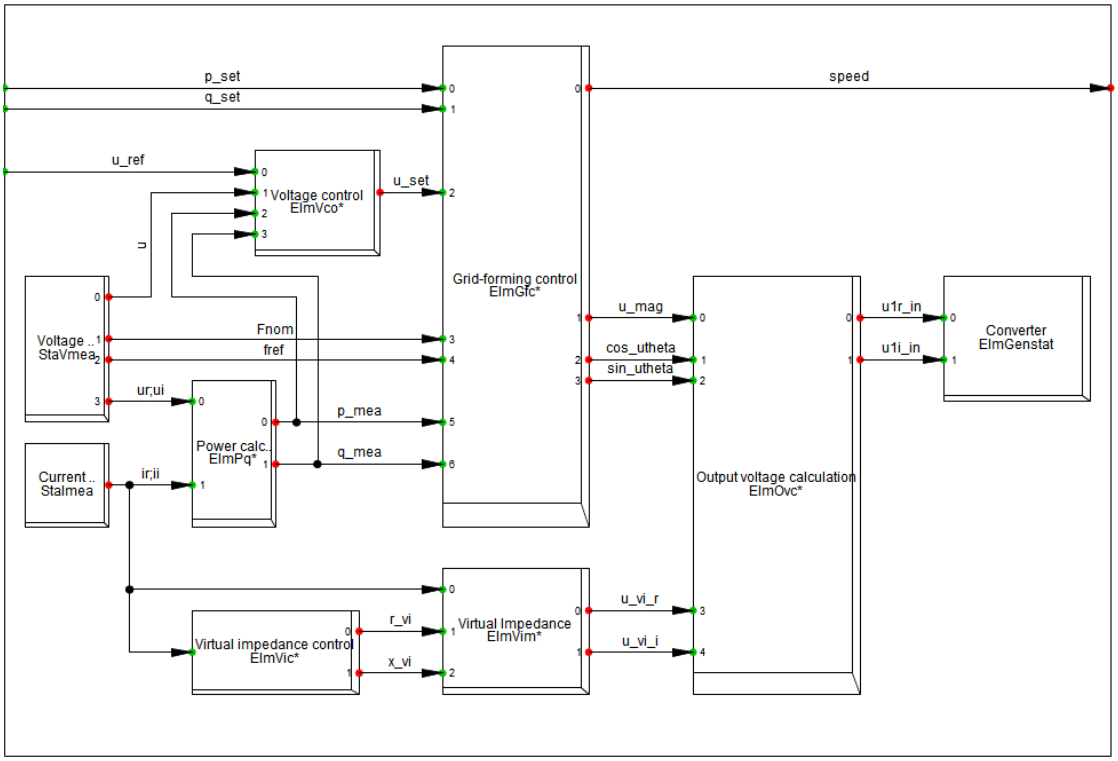


Figure 2.1: Composite Model of the Virtual Synchronous Machine.

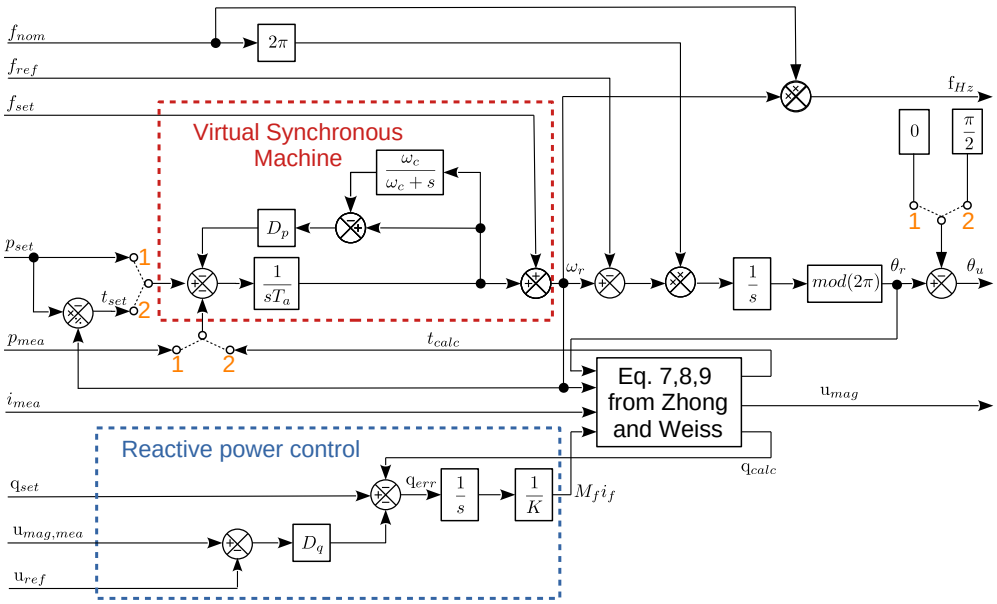


Figure 2.2: Block diagram of VSM and Synchronverter.

torque t_{calc}, stator voltage magnitude $u_{s,mag}$ and the reactive power q_{calc} as follows:

$$t_{calc} = M_f i_f (i_{a,mea} sin(\theta_r) + i_{b,mea} sin(\theta_r - \frac{2\pi}{3}) + i_{c,mea} sin(\theta_r - \frac{4\pi}{3})), \quad (4)$$

$$u_{s,mag} = \dot{\theta}_r M_f i_f, \quad (5)$$

$$q_{calc} = -\dot{\theta}_r M_f i_f (i_{a,mea} cos(\theta_r) + i_{b,mea} cos(\theta_r - \frac{2\pi}{3}) + i_{c,mea} cos(\theta_r - \frac{4\pi}{3})), \quad (6)$$

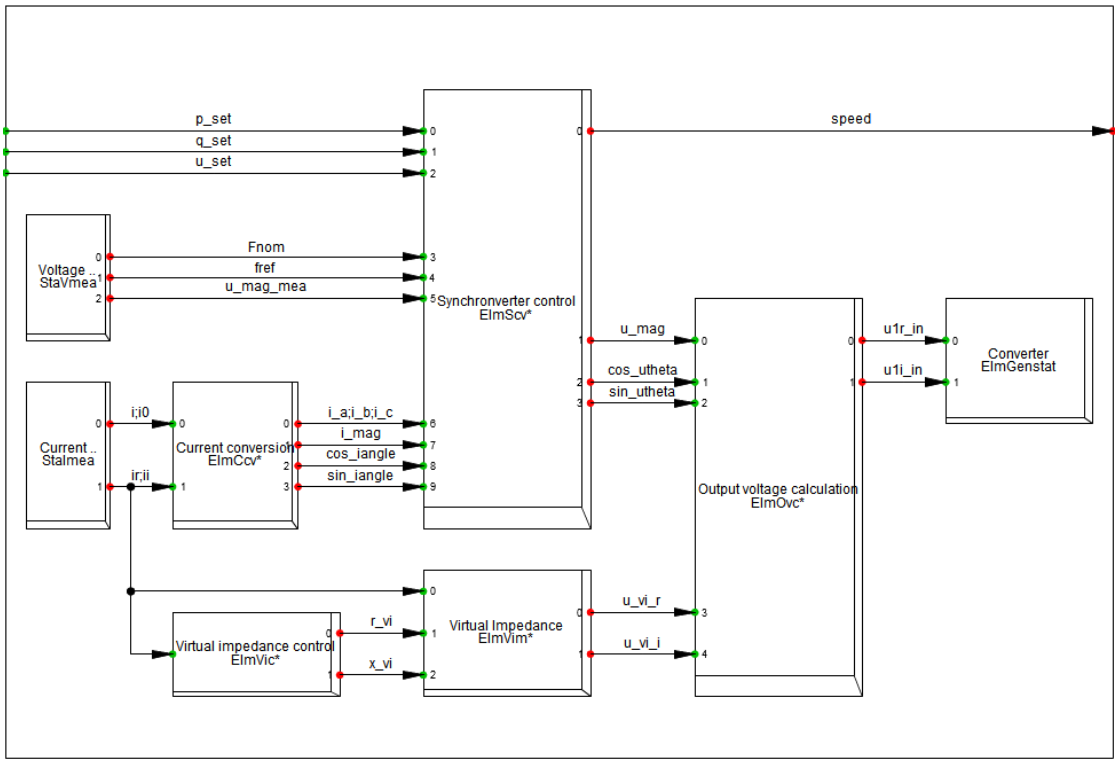


Figure 2.3: Composite Model of the Synchronverter.

where $M_f i_f$ is the simplified excitation, $i_{abc,mea}$ is the measured output current and θ_r is the rotor angle.
The reactive power controller is depicted in the lower left part of Fig. 2.2. It consist of an integral controller with the gain $\frac{1}{K}$. The reactive power error q_{err} depends on the set value q_{set}, the calculated value q_{calc} and the voltage droop as follows:

$$q_{err} = q_{set} - q_{calc} - D_q(u_{mag,mea} - u_{set}), \tag{7}$$

where $u_{mag,mea}$ is the measured voltage magnitude, u_{set} is the the voltage set point and D_q is the voltage droop coefficient.
The stator voltage θ_u is proportional to the change of flux induced by the rotor field and, therefore, lags the rotor angle θ_r by $\pi/2$.

2.3 Droop Control

The droop control is based on the same frame as the VSM (Fig. 2.1). In the grid-forming droop control, the frequency $\Delta\omega_{droop}$ and voltage magnitude Δu_{droop} deviations from the load flow initialization are calculated according to the droop equations [2]:

$$\Delta\omega_{droop} = m_p \Delta p_{LPF}, \tag{8}$$
$$\Delta v_{droop} = m_q \Delta q_{LPF}, \tag{9}$$

where m_p and m_q are the active and reactive power droop coefficients and Δp_{LPF} and Δq_{LPF} are the low-pass filtered active and reactive power deviations from the load flow initialization, respectively.
The parameters f_{set}, p_{set}, u_{set} and q_{set} are usually initialized according to the initial load flow. However, they are modelled as input variables to the droop controller and, therefore, can be

controlled from outside the block.
It was shown in [1] that the frequency calculation of the droop control and VSM are actually similar when parameters are tuned accordingly.

2.4 Virtual Impedance

The real and reactive voltage drop over an algebraic type virtual impedance ia calculated as follows:

$$\Delta u_{vi,r} = r_{vi} i_{vi,r} - x_{vi} i_{vi,i}, \tag{10}$$

$$\Delta u_{vi,i} = r_{vi} i_{vi,i} + x_{vi} i_{vi,r}, \tag{11}$$

where r_{vi} is the resistance and x_{vi} the reactance of the virtual impedance. The current i_{vi} is the current supplied by the converter.
The virtual impedance controller can be used to adjust the virtual impedance parameters to the grid condition. A typical application would be to increase the virtual impedance during short-circuit events. The virtual impedance is adjusted in proportion to the low-pass filtered current absolute value $|i_{vi}|$ when a certain threshold i_{lim} is exceeded [3] as given in (12),

$$z_{vi} = \begin{cases} r_{vi} + jx_{vi}, & \text{if } |i_{vi}| \leq i_{lim} \\ (k_{pr} r_{vi} + jk_{px} x_{vi})(|i_{vi}| - i_{lim}) + r_{vi} + jx_{vi} & \text{otherwise,} \end{cases} \tag{12}$$

where k_{pr} and k_{px} are the proportional factors for the virtual resistor and virtual reactance, respectively.

2.5 Output Voltage Calculation

In the block 'Output voltage calculation', the voltage of the grid-forming controller and the voltage drop of the virtual impedance are used to calculate the final output voltage of the converter.
In addition to the current limitation by the virtual impedance, another mechanism to limit the output current and to avoid overloading of the converter is implemented in this block. At first, the voltage drop over the converter series impedance $\underline{u}_{drop,unlim}$ is calculated:

$$\underline{u}_{drop,unlim} = \underline{u}_{con,unlim} - \underline{u}_{term}, \tag{13}$$

where $\underline{u}_{con,unlim}$ is the unlimited converter voltage and $\underline{u}_{term}$ is the voltage of the connected terminal. It is known in advance that the absolute value of this voltage drop should not exceed a certain value $u_{drop,max}$ given by the converter series impedance $\underline{z}_{series}$ and the maximum converter current $i_{con,lim}$:

$$u_{drop,max} = |\underline{z}_{series}| i_{con,lim}. \tag{14}$$

Therefore, the actual absolute value of the voltage drop $|\underline{u}_{drop,lim}|$ is limited by $u_{drop,max}$ (while the angle of $\underline{u}_{drop,lim}$ remains the same as the angle of $\underline{u}_{drop,unlim}$). Finally, the limited converter voltage $\underline{u}_{con,lim}$ is derived by adding the limited voltage $\underline{u}_{drop,lim}$ drop to the terminal voltage $\underline{u}_{term}$:

$$\underline{u}_{con,lim} = \underline{u}_{drop,lim} + \underline{u}_{term}. \tag{15}$$

3 RMS-Simulation

With respect to RMS simulation, the described templates / models of grid-forming converters implemented so far, are only valid for balanced RMS simulation. For RMS simulation a small

step size for electromechanical transients of 0.5 ms might be required during dynamic events in some cases.

3.1 Virtual Synchronous Machine

The model as described in Sec. 2.1 is used for the VSM.

3.2 Synchronverter

DSL uses symmetric components transformed into the reference frame of the reference frequency in RMS-simulations. The active and reactive power in symmetrical components is calculated as follows:

$$p = u_0 i_0 cos(\theta_{u,0} - \theta_{i,0}) + u_1 i_1 cos(\theta_{u,1} - \theta_{i,1}) + u_2 i_2 cos(\theta_{u,2} - \theta_{i,2}), \tag{16}$$

$$q = u_0 i_0 sin(\theta_{u,0} - \theta_{i,0}) + u_1 i_1 sin(\theta_{u,1} - \theta_{i,1}) + u_2 i_2 sin(\theta_{u,2} - \theta_{i,2}), \tag{17}$$

where u_{012} are the magnitudes of the symmetrical components of the voltage with their angles $\theta_{u,012}$ and where i_{012} are the magnitudes of the symmetrical components of the current with their angles $\theta_{i,012}$.
The voltage output of a Synchronverter is always symmetric. Therefore, only the positive sequence is relevant which leads to

$$p = u_1 i_1 cos(\theta_{u,1} - \theta_{i,1}), \tag{18}$$

$$q = u_1 i_1 sin(\theta_{u,1} - \theta_{i,1}). \tag{19}$$

For the Synchronverter equations (4), (5) and (6), the rotor angle θ_r is used instead of the voltage angle $\theta_{u,1}$. It lags the voltage angle by $90°$. Assuming that the frequency is close to its nominal value, the torque t_{calc} and reactive power q_{calc} are then written as

$$t_{calc} = M_f i_f i_{mea,1} sin(\theta_r - \theta_{i,mea,1}), \tag{20}$$

$$q_{calc} = -\dot{\theta} M_f i_f i_{mea,1} cos(\theta_r - \theta_{i,mea,1}). \tag{21}$$

3.3 Droop Control

The droop equations as elaborated in Sec. 2.3 are implemented.

3.4 Virtual Impedance

The implementation as depicted in Sec. 2.4 is used.

3.5 Output Voltage Calculation

The implementation as described in Sec. 2.5 is used.

4 EMT-Simulation

For use in EMT simulation, it is recommended to disable the solver option ”Fast convergence check” in calculation of initial conditions when using the templates / models of grid-forming converters.

4.1 Virtual Synchronous Machine

The model as described in Sec. 2.1 is used for the VSM. The reference angle θ_{ref} is only relevant in RMS-simulations and is set to zero in EMT simulations.

4.2 Synchronverter

The reference angle θ_{ref} is only relevant in RMS-simulations and is set to zero in EMT simulations.
Equations (4), (5) and (6) hold for the Synchronverter in EMT simulations, except for the factor $2/3$. The factor considers the amplitude invariant frame transformation used in DSL (instead of power invariant):

$$t_{calc} = 2/3\ M_f i_f (i_a sin(\theta_r) + i_b sin(\theta_r - \frac{2\pi}{3}) + i_c sin(\theta_r - \frac{4\pi}{3})) \tag{22}$$

$$u_{s,mag} = \dot{\theta_r} M_f i_f \tag{23}$$

$$q_{calc} = -2/3\ \dot{\theta_r} M_f i_f (i_a cos(\theta_r) + i_b cos(\theta_r - \frac{2\pi}{3}) + i_c cos(\theta_r - \frac{4\pi}{3})) \tag{24}$$

4.3 Droop Control

The droop equations as elaborated in Sec. 2.3 are implemented. The reference angle θ_{ref} is only relevant in RMS-simulations and is set to zero in EMT simulations.

4.4 Virtual Impedance

The implementation as depicted in Sec. 2.4 is used.

4.5 Output Voltage Calculation

The implementation as described in Sec. 2.5 is used.

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